



Deiðer a dúnitars.





Necessary, not sufficient; more cheap

unitary matrix $S::SL=RS$



$$\text{tr}(L) = \text{tr}(R)$$



Necessary and sufficient; expensive

$$\exists \text{ unitary matrix } S:: SL = RS$$



$$\text{eig}(L) = \text{eig}(R)$$



Idea: Trace eliminate bad candidates early!

Decide whether a unitary S exists.

FOR UNITARY CIRCUITS L AND R

EIGENVALUE TEST · Lemma 5.3

Necessary and sufficient; expensive

$$\exists \text{ unitary matrix } S : SL = RS \quad \Leftrightarrow \quad \text{eig}(L) = \text{eig}(R)$$

TRACE FILTER · Lemma 5.4

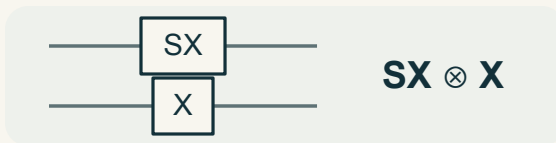
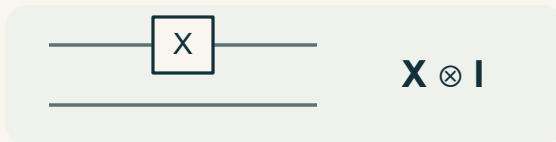
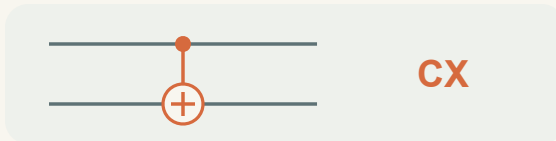
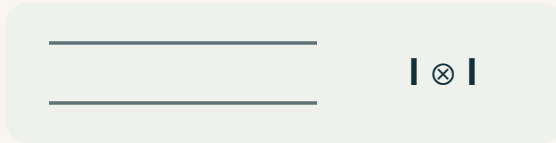
Necessary, not sufficient; more cheap

$$\exists \text{ unitary matrix } S : SL = RS \quad \Rightarrow \quad \text{tr}(L) = \text{tr}(R)$$

Idea: Use Trace to eliminate bad candidates early!

Group by trace, then solve for S

1 Concrete terms



2 qubits · 10 pairs